MaiaEdge

TECHNICAL REFERENCE

MaiaEdge Three-POP Deployment

SIPNAV
Reference Architecture.

Private path architecture for SIPNAV's three-POP inter-exchange voice carrier network.
Prepared for Ken Rice, Head of Engineering.

Private paths. Any network. Instantly.

Carrier infrastructure for federated private networking

01 · ENVIRONMENT

Your Environment

SIPNAV operates one of the highest-throughput voice carrier networks in the market: roughly one billion phone calls per day, with peaks of 50,000 to 75,000 new calls per second. Three POPs anchor the infrastructure.

POP	Facility	Key Connectivity
Los Angeles	One Wilshire (primary) · Digital Realty (adjacent)	Multiple 40G transit; Equinix one floor above via cross-connect
Dallas	60 Hudson + DataMark / Infomark	Multiple 40G transit
Tampa	DataMark / Infomark footprint	Dark fiber to adjacent buildings via Fiber Light

Routing topology: Three separate autonomous systems (West, Central, East) for disaster recovery. Juniper MX-based forwarding at every site.

Current DR hierarchy: Wave ring (primary) › VPLS › GRE tunnels (last resort). Goal: eliminate GRE entirely.

Traffic profile: Two distinct flow types run simultaneously. Jumbo frames carry SIP signaling metadata; G.711 audio flows at 250-byte packets at approximately 100 packets per second per call. At 500,000 concurrent calls, audio volume alone is approximately 50 Gbps. SIPNAV deliberately keeps media off the WAN wherever possible.

02 · PROBLEMS / SOLUTIONS

Problems We Are Solving

45 to 50 fiber cuts in 90 days. Long-haul runs have been unreliable. When the wave ring fails, the path degrades to VPLS, then to GRE over DIA as last resort. GRE is unencrypted, adds overhead, and introduces latency variance that degrades call quality. MaiaEdge replaces GRE with encrypted, deterministic PBC paths over existing DIA so the fallback is indistinguishable from the primary.

Verizon IPsec tunnel proliferation. Dozens of virtualized IPsec appliances for Verizon SIP signaling peering. Each virtual box is an operational liability: patching, monitoring, capacity planning, and failure recovery. MaiaEdge replaces the fleet with a single PCE-managed private path per interconnect site.

LA cross-connect economics. Digital Realty and Equinix sit one floor apart, but a physical cross-connect is priced prohibitively. A MaiaEdge PBC-over-DIA path creates a Layer 2 private connection between the buildings without the cross-connect charge.

Voice carrier marketplace access. Finding and onboarding peering partners — Verizon, Windstream, TSI-tier providers — requires manual NNI setup per carrier. The MaiaEdge marketplace surfaces peering-ready operators and cuts onboarding from weeks to minutes.

03 · ARCHITECTURE

Reference Architecture

One PBC at each POP. All three register to a single cloud-hosted PCE for path computation, policy enforcement, and telemetry aggregation. No BGP, OSPF, or MPLS is added to the existing Juniper MX environment. MaiaEdge operates as an overlay, consuming existing DIA at each POP as transport for the private path fabric.

Los Angeles · One Wilshire

West anchor. The PBC connects to existing 40G transit DIA and establishes encrypted private paths to Dallas and Tampa. A second path to Equinix LA (one floor up) runs PBC-over-DIA: a Layer 2 private path across the building boundary without a physical cross-connect charge. Equinix cloud on-ramp access (AWS Direct Connect, Azure ExpressRoute) is available through this path at SIPNAV's option.

Dallas · 60 Hudson + DataMark

Central AS anchor. The PBC at 60 Hudson establishes encrypted paths to both coasts. DataMark provides the secondary footprint; an HA unit at DataMark completes intra-Dallas redundancy. The PCE treats both Dallas units as a single logical site for path computation.

Tampa · Fiber Light Dark Fiber

East AS anchor. The Fiber Light dark fiber between buildings is MaiaEdge-native transport: PBCs terminate the fiber at each end and deliver a fully encrypted, telemetry-visible Layer 2 path. This replaces raw unencrypted dark fiber use and adds hop-by-hop visibility that was previously unavailable.

Backup Path Architecture · Replacing GRE

Layer	Current	With MaiaEdge
Primary	Wave ring	Wave ring (unchanged)
Secondary	VPLS	VPLS (unchanged)
Tertiary (last resort)	GRE over DIA — unencrypted, high overhead	PBC private path over DIA — AES-256-GCM encrypted, <2 μs overhead, full telemetry

The PBC private path uses the same DIA circuits that currently carry GRE — no new transport is required. Failover is policy-driven by the PCE: when it detects path degradation above threshold, it instantiates the PBC path automatically. From the Juniper MX perspective, the MaiaEdge path looks like any other L2 circuit. No routing protocol changes are needed.

04 · TRAFFIC PROFILE

Traffic Profile Considerations

SIPNAV's dual traffic profile requires specific treatment on MaiaEdge paths.

Traffic Type	Characteristics	MaiaEdge Handling
SIP signaling	Jumbo frames, 9,000-byte MTU	PBC-over-DIA handles natively. PCE path selection excludes hops that cannot support jumbo frames from signaling route candidates.
G.711 audio	250-byte packets, 100 pps per call. ~50 Gbps at 500K concurrent calls. Jitter tolerance ~20 ms per call.	Deterministic path engineering — fixed hop selection, no dynamic rerouting under load — keeps jitter bounded. PCE monitors per-path jitter in real time and triggers path switch before the jitter budget is exhausted.
Media locality	SIPNAV policy: keep media off WAN where possible	Compatible. MaiaEdge carries signaling and control across POPs. PCE can be policy-gated to carry media only when intra-POP handling is unavailable.

05 · SIGNALING SOLUTION

Verizon Signaling Solution

SIPNAV currently maintains dozens of virtualized IPsec boxes solely for Verizon SIP signaling peering. Each box represents provisioning overhead, a patching surface, and a failure domain.

The MaiaEdge alternative: a single PBC at each SIPNAV POP federates with Verizon's MaiaEdge endpoint. SIP signaling peering runs as an encrypted Layer 2 private path over existing DIA. One PBC per POP replaces the entire fleet of virtual IPsec appliances for Verizon peering.

Today	With MaiaEdge
Dozens of virtualized IPsec appliances	Single PBC per POP — PCE-managed private path
Per-peer IPsec configuration for each carrier	One path per interconnect site, policy-defined in PCE
Virtual appliance patching cycles + capacity planning	Software-managed subscription — no per-appliance patching
Manual troubleshooting when a specific tunnel degrades	Hop-by-hop telemetry (jitter, loss, latency) on every path
No logical isolation between signaling peers on shared hardware	Q-in-Q tagging ensures per-carrier traffic isolation on the same PBC

06 · MARKETPLACE

Marketplace and Carrier Peering

The MaiaEdge marketplace surfaces operators who have deployed PBCs and are available for NNI peering. For SIPNAV, the primary targets are Tier 1 voice carriers (Verizon, Windstream) and TSI-scale aggregators.

Step	Action	Who Does It
1	Partner appears as peering-ready node in PCE portal	Automatic — marketplace discovery
2	Select partner, define path policy (bandwidth, redundancy, signaling scope)	SIPNAV engineering

Step	Action	Who Does It
3	PCE instantiates the path	Automatic — no LOA, no BGP session, no out-of-band VLAN coordination
4	Telemetry active immediately: jitter, loss, latency per hop	Automatic

> CyrusOne near-term peering candidate. CyrusOne has been flagged as a relevant near-term opportunity given its carrier density. A PBC at CyrusOne (Dallas or shared footprint) allows SIPNAV to reach any CyrusOne-connected carrier through the marketplace path without a direct per-carrier NNI build.

07 · WHAT YOU REPLACE

What MaiaEdge Replaces

Today	With MaiaEdge
GRE tunnels as last-resort failover	Encrypted PBC private paths over DIA — same transport, zero GRE overhead
Dozens of virtualized Verizon IPsec appliances	Single PBC per POP with PCE-managed signaling paths
Manual NNI setup per carrier peering partner	Marketplace-driven: select partner, define policy, path is live
No telemetry across DIA fallback paths	Hop-by-hop jitter, loss, latency across every path including fallback
LA cross-connect charge to reach Equinix	PBC-over-DIA between buildings — Layer 2 path without the cross-connect fee
Tampa dark fiber used as raw unencrypted transport	PBC-terminated fiber with AES-256-GCM encryption and full telemetry

08 · PRICING

Subscription and Pricing

MaiaEdge is sold as a combined hardware and orchestration software subscription per PBC unit. The PCE is cloud-hosted and included in the subscription. Monthly invoicing available.

Term	Monthly Rate Per PBC	Notes
1-year	~$1,900 / month	Monthly invoicing available
3-year	~$1,600–$1,700 / month	Monthly invoicing available

Item	SIPNAV Deployment
Base units	3 PBCs — one per POP (LA One Wilshire, Dallas 60 Hudson, Tampa)
Throughput tier	100G — appropriate given 40G transit capacity and ~50G peak audio volume
Optional HA units	Dallas DataMark secondary · Tampa building 2 — priced at discounted rate

Item	SIPNAV Deployment
PCE orchestration	Cloud-hosted, included. Covers all PBCs under one control plane.
Peering / marketplace	No per-path or per-peering-partner charges. Marketplace access included.